

CSCE 649 PHYSICALLY-BASED MODELING · MODULE 02 · LECTURE NOTES

Numerical Integration

Why your simulation explodes, and which scheme stops it.

MODULE OUTCOMES

- Derive explicit Euler and explain why it gains energy.
- Distinguish accuracy from stability, and order from behaviour.
- Explain what symplectic means and why it matters over long runs.
- Define stiffness and explain why it forces implicit integration.
- Choose an integrator from a system's requirements.

1 Explicit Euler, and exactly how it fails

$$\mathbf{x}(\mathbf{t} + \Delta t) = \mathbf{x}(\mathbf{t}) + \Delta t \cdot \mathbf{f}(\mathbf{x}(\mathbf{t}), \mathbf{t})$$

Evaluate the derivative where you currently are, and step along it. It is the first method anyone writes, it is first-order accurate — local error $O(\Delta t^2)$, global error $O(\Delta t)$ — and it is unstable for oscillatory systems regardless of step size.

THE GEOMETRIC REASON IT GAINS ENERGY

Take a particle in a circular orbit under a central force. The true trajectory curves continuously toward the centre.

Explicit Euler steps along the tangent line, because that is what the derivative at the current point describes. **A tangent to a circle always lies outside the circle.**

So after each step the particle is slightly further from the centre than it should be, and moving slightly too fast for its new radius. **Total energy increases, monotonically, every single step.** Over thousands of steps the orbit spirals outward and eventually escapes.

Reducing the time step does not fix this. It slows the growth — halving Δt roughly halves the rate — but the sign of the error is always the same, so the energy still grows without bound. **The error is systematic, not random, and that is the entire lesson:** no amount of refinement rescues a method whose error does not cancel.

	Accuracy (order)	Stability
Asks	How fast does error shrink as Δt shrinks?	Does error grow as steps accumulate?
Measured by	The exponent: Euler is $O(\Delta t)$, RK4 is $O(\Delta t^4)$.	The largest Δt for which the system does not diverge.

	Accuracy (order)	Stability
Matters over	A few steps.	Thousands of steps.
Relationship	Independent. A fourth-order method can be unstable; a first-order method can be perfectly stable.	This is what you actually care about in a simulator, which runs for hours.

Conflating these two is the most common conceptual error in this module. Someone whose simulation explodes reaches for RK4, because it is 'more accurate'. RK4 is not symplectic and will also lose energy over long runs; it buys accuracy, which was not the problem.

2 Symplectic integrators

```
// Explicit (forward) Euler -- unstable for oscillators
p += v * dt;           // uses OLD v
v += a * dt;

// Symplectic (semi-implicit) Euler -- stable
v += a * dt;           // update velocity FIRST
p += v * dt;           // then use the NEW v
```

The two lines are swapped. The cost is identical, the order of accuracy is identical, and the long-term behaviour is completely different: energy now oscillates within a bounded band instead of growing without limit. **If you take one practical result from this course, it is this one.**

WHAT 'SYMPLECTIC' MEANS AND DOES NOT MEAN

A symplectic integrator preserves the **phase-space volume** of the system — a geometric structure that Hamiltonian mechanics shows the true dynamics preserves exactly. Liouville's theorem is the formal statement.

The practical consequence is that the energy error is bounded for all time. It oscillates with the system's period; it does not accumulate. A symplectic simulation of the solar system can run for millions of years without the planets drifting away.

It does not mean energy is conserved exactly. What the method conserves exactly is the energy of a slightly *different* system — a 'shadow Hamiltonian' close to the true one. That is why the error oscillates rather than vanishing.

This is why a symplectic first-order method outperforms a non-symplectic fourth-order one over long runs. A qualitative structural property beats three orders of accuracy, which is a genuinely surprising result and one of the more satisfying facts in numerical analysis.

Method	Order	Symplectic	Cost per step	Use for
Explicit Euler	1	No	1 derivative	Nothing in production. Teaching, and as a baseline for the energy plot.
Symplectic Euler	1	Yes	1 derivative	The sensible default. Games, real-time, anything with contact.
Velocity Verlet	2	Yes	1 derivative	Strictly better than symplectic Euler for the same cost when forces do not depend on velocity. Molecular dynamics uses it universally.
Midpoint / RK2	2	No	2 derivatives	Smooth non-oscillatory systems.
RK4	4	No	4 derivatives	High accuracy over short runs. Compare it against four symplectic Euler steps , not against one — that is the fair comparison, and it often loses.

3 Stiffness

WHAT STIFFNESS IS

A system is **stiff** when it contains processes whose characteristic timescales differ by orders of magnitude. Cloth is the canonical example. The stretch springs that stop the fabric from extending are very strong, so they oscillate at kilohertz frequencies. The cloth's visible motion — draping, swinging — happens at a few hertz.

An explicit integrator must resolve the fastest timescale present, or it goes unstable, even though nothing visually interesting happens at that frequency. The method cannot ignore a mode simply because we are uninterested in it.

So Δt is dictated by the stiffest spring in the mesh, which may be thousands of times smaller than the visible motion requires. Stiffer cloth means smaller steps means a slower simulation — and this is a property of the method, not of the hardware.

$$\Delta t < 2 \sqrt{m / k}$$

For a mass-spring system with explicit integration, the step is bounded by the stiffest spring constant k and the smallest mass m . **Making a material a hundred times stiffer requires a ten times smaller step**, so a hundred times the work for the same simulated duration. This is a hard limit derived from the method's stability region, not a tuning guideline, and it is worth measuring empirically — the predicted threshold matches the observed one closely.

3.1 Implicit integration

$$\mathbf{x}(t + \Delta t) = \mathbf{x}(t) + \Delta t \cdot \mathbf{f}(\mathbf{x}(t + \Delta t))$$

Backward Euler evaluates the derivative at the destination rather than the origin. The unknown now appears on both sides, so each step requires solving a system of equations rather than evaluating an expression. Linearising about the current state gives:

$$(\mathbf{I} - \Delta t \cdot \partial \mathbf{f} / \partial \mathbf{x}) \Delta \mathbf{x} = \Delta t \cdot \mathbf{f}(\mathbf{x})$$

A sparse linear system, solved each step — typically by conjugate gradient, since the matrix is large, sparse, and usually symmetric positive definite. The Jacobian $\partial \mathbf{f} / \partial \mathbf{x}$ must be assembled, which is the fiddly part.

IMPLICIT METHODS BUY STABILITY WITH DISSIPATION

Backward Euler is unconditionally stable: it will not diverge at *any* time step. Baraff and Witkin's 1998 cloth paper applied exactly this and changed what was possible — cloth that had required thousands of tiny explicit steps could now be advanced one frame at a time.

But it is strongly dissipative. It removes energy aggressively, and at large time steps the result is visibly over-damped: cloth that moves as though through treacle, losing the high-frequency flutter that makes fabric look like fabric.

The damping is not a defect to be fixed — it is the mechanism. The method is stable precisely because it drives high-frequency modes to zero, and those modes include ones you might have wanted.

Less dissipative implicit schemes exist — implicit midpoint, BDF2 — and are harder to make robust in the presence of contact and collision. Most production cloth uses backward Euler and compensates artistically for the damping.

4 Choosing, and what matters more

System	Integrator	Reasoning
Particles, loose springs, general dynamics.	Symplectic Euler.	Cheap, stable for non-stiff systems, bounded energy. The right default.
Rigid bodies with contact.	Symplectic Euler with substepping.	Contact resolution requires small steps regardless, so a higher-order method buys nothing.
Stiff cloth.	Implicit (backward Euler).	Explicit integration is unaffordable by §3's bound.
Orbital mechanics, molecular dynamics, very long runs.	Velocity Verlet.	Symplectic, second order, one derivative evaluation. Energy bounded over millions of steps.
High accuracy over a short interval.	RK4.	Highest order per step — but compare against four symplectic steps for fairness.
Games, interactive, robustness above all.	XPBD (Module 04).	Cannot explode under any circumstances , and the stiffness is directly tunable. Theoretically weak and practically dominant.

- **Substepping is the most effective practical lever here.** Taking four small symplectic Euler steps per frame is usually better, simpler, and more robust than one large step with a sophisticated method — and it composes with everything else.
- **Use a fixed time step.** A step tied to real frame time makes the simulation's behaviour depend on the machine's performance and destroys determinism (Module 01). Module 13 covers decoupling simulation rate from render rate.
- **Clamp velocities** as a safety net. It is unphysical and it prevents one bad frame — an interpenetration, a degenerate contact — from destroying an otherwise good simulation.

- **Assert on NaN every step, in debug builds.** A single NaN propagates through the entire state within a few steps, at which point the original cause is unrecoverable. Catching it on the step it appears turns an impossible bug into an easy one.
- **Measure rather than assume.** Plot the energy. The curve will tell you whether the integrator is behaving as the theory says it should, and it is the only way to know.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Baraff & Witkin — Differential Equation Basics and Implicit Methods (free notes)	Sections B and E of the course pack. The implicit cloth derivation of §3 is in the 1998 paper linked from the same page.
Ten Minute Physics — integration and stability episodes	The energy behaviour of §1 and §2, demonstrated interactively — which is far more convincing than reading about it.
Hairer, Lubich & Wanner — Geometric Numerical Integration	The authoritative account of why symplectic methods behave as they do. Chapter 1 is free and its figures are the best illustration of §2.
Baraff & Witkin — Large Steps in Cloth Simulation (1998, free)	The paper that made implicit cloth practical. Read it before Module 10.
Erin Catto — Numerical Integration (GDC, free slides)	The game-development view: why symplectic Euler and substepping beat cleverness in practice.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Implement explicit Euler, symplectic Euler, velocity Verlet, and RK4 behind the interface from Module 01.
- 2 **Produce the energy plot:** a particle in a circular orbit under each integrator, 10,000 steps, total energy against time on one figure. Explain the shape of each curve. This figure belongs in your portfolio.
- 3 Halve the time step for explicit Euler and show that the energy still grows, only more slowly. Report the relationship.
- 4 Measure the stability limit for a single mass-spring system experimentally and compare against $2\sqrt{m/k}$. Report the agreement.
- 5 Vary the spring constant over three orders of magnitude and plot the maximum stable time step against it on log-log axes. The slope should be -0.5 .

- 6 Compare RK4 at Δt against symplectic Euler at $\Delta t/4$ — equal derivative evaluations. Report accuracy and energy behaviour for both.
- 7 Implement backward Euler for a mass-spring system. Confirm it is stable at a time step a hundred times larger than the explicit limit, and measure the energy it removes.
- 8 Build a stiff cloth patch and find the explicit stability limit empirically. Estimate the cost of simulating one second of it.
- 9 Introduce a NaN deliberately and count how many steps until the entire state is corrupted. Then add the assertion.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Write explicit Euler and explain geometrically why it gains energy on an orbit.
- 2 Why does reducing the time step not make explicit Euler stable?
- 3 Distinguish accuracy from stability, and say which matters for a simulator.
- 4 What is the difference between explicit and symplectic Euler, and what does it change?
- 5 What does symplectic mean, and what does it *not* guarantee?
- 6 Why can a symplectic first-order method beat a non-symplectic fourth-order one?
- 7 Define stiffness and explain why it limits explicit time steps.
- 8 State the explicit stability bound and say what happens when stiffness increases a hundredfold.
- 9 Why are implicit methods unconditionally stable, and what is the cost?
- 10 Name three practical measures that matter more than the choice of integrator.